

ST2131 MT Cheatsheet AY2627 S1

1 Combinatorial Analysis

Three basic principles
(AP) **Addition:** $E_1, \dots, E_k$ pairwise disjoint $\Rightarrow$ #ways at least one occurs $= \sum n_i$. Sets: $A_i \cap A_j = \emptyset$ ($i \neq j$) $\Rightarrow |\bigcup A_i| = \sum |A_i|$.
(MP) **Multiplication:** task = sequence of r stages with $n_1, \dots, n_r$ choices $\Rightarrow \prod n_i$. Sets: $|A_1 \times \dots \times A_r| = \prod |A_i|$.
(CP) **Complement:** $A \subseteq U \Rightarrow |U \setminus A| = |U| - |A|$.

AP = split into disjoint cases (“or”); MP = build step by step (“and then”); CP = count the bad ones when “good” is messy (e.g. $a \neq b$: count all, subtract $a = b$).

Examples: #functions $\{1..n\} \rightarrow \{0,1\} = 2^n$; # k -ary sequences of length $n = k^n$; $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$.

Permutations and combinations
 r -permutation (ordered, no repeats): $P_r^n = n(n-1) \cdots (n-r+1) = \frac{n!}{(n-r)!}$; $0! = 1$, $P_n^n = n!$.

r -combination (subset): $\binom{n}{r} = C_r^n = \frac{P_r^n}{r!} = \frac{n!}{r!(n-r)!}$.

Permutation = choose a subset, then arrange it: $P_r^n = C_r^n \cdot r!$. Dividing by $r!$ “forgets” the order.

Symmetry: $\binom{n}{r} = \binom{n}{n-r}$ (choosing who’s in = choosing who’s out).
Pascal: $\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$. *Combinatorial proof:* does the subset contain element 1 or not?
With repetition: r -permutations of n objects $= n^r$.
Multiset permutations: r objects, r_i of type i :

$$P(r; r_1, \dots, r_n) = \frac{r!}{r_1! r_2! \cdots r_n!}$$

(label identical copies, then divide out the $r_i!$ overcount). E.g. $aaabc$: $5!/3! = 20$.

- Standard tricks
- Blocks together:** arrange blocks, then within blocks. 4 math, 3 chem, 2 hist, 1 lang books grouped: $4! \cdot 4! 3! 2! 1! = 6912$.
 - Gap method (no two adjacent):** line up the $n - m$ “good” items, giving $n - m + 1$ gaps (incl. ends); pick m : $\binom{n-m+1}{m}$.
 - Restriction by complement:** committees of 3 from 7 men avoiding 2 feuding men: $\binom{7}{3} - \binom{2}{2} \binom{5}{1} = 30$.
 - Separate rankings: 6 men, 4 women ranked among themselves: $6! 4!$.

Binomial theorem

$$(x + y)^n = \sum_{r=0}^n \binom{n}{r} x^{n-r} y^r$$

Why: from n factors $(x + y)$, choose which r factors give y : $\binom{n}{r}$ ways. (Also provable by induction + Pascal.)

- $x = y = 1$: $\sum_r \binom{n}{r} = 2^n = \text{\#subsets of an } n\text{-set (each element in/out). Nonempty: } 2^n - 1$.
- $x = 1, y = -1$: $\sum_r (-1)^r \binom{n}{r} = 0$ ($n \geq 1$).
- Hence $\binom{n}{0} + \binom{n}{2} + \cdots = \binom{n}{1} + \binom{n}{3} + \cdots = 2^{n-1}$.

Multinomial coefficients

#ways to put n **distinct** objects into m **distinct** boxes with n_i in box i ($\sum n_i = n$):

$$\binom{n}{n_1, \dots, n_m} = \binom{n}{n_1} \binom{n - n_1}{n_2} \cdots = \frac{n!}{n_1! \cdots n_m!}$$

Multinomial Thm:

$$(x_1 + \cdots + x_m)^n = \sum_{n_1 + \cdots + n_m = n} \binom{n}{n_1, \dots, n_m} x_1^{n_1} \cdots x_m^{n_m}$$

2 Axioms of Probability

Sample space and events
 S = set of all outcomes; event $E \subseteq S$. $E \cup F$ (at least one), $E \cap F = EF$ (both), E^c (not E), $S^c = \emptyset$. **Mutually exclusive:** $EF = \emptyset$. $E \subseteq F$: E occurs $\Rightarrow F$ occurs.
Commutative, associative, distributive laws: $(E \cup F)G = EG \cup FG$, $EF \cup G = (E \cup G)(F \cup G)$.
De Morgan: $(\bigcup E_i)^c = \bigcap E_i^c$, $(\bigcap E_i)^c = \bigcup E_i^c$.

The three axioms
A1: $0 \leq \mathbb{P}(E) \leq 1$. **A2:** $\mathbb{P}(S) = 1$.
A3: $E_1, E_2, \dots$ mutually exclusive $\Rightarrow \mathbb{P}(\bigcup_{i=1}^\infty E_i) = \sum_{i=1}^\infty \mathbb{P}(E_i)$.
Consequences: $\mathbb{P}(\emptyset) = 0$; finite additivity (pad with $\emptyset$'s).

Simple propositions

- $\mathbb{P}(E^c) = 1 - \mathbb{P}(E)$.
- $E \subseteq F \Rightarrow \mathbb{P}(E) \leq \mathbb{P}(F)$ (write $F = E \cup E^c F$).
- $\mathbb{P}(E \cup F) = \mathbb{P}(E) + \mathbb{P}(F) - \mathbb{P}(EF)$.
- $\mathbb{P}(E \cup F \cup G) = \mathbb{P}(E) + \mathbb{P}(F) + \mathbb{P}(G) - \mathbb{P}(EF) - \mathbb{P}(EG) - \mathbb{P}(FG) + \mathbb{P}(EFG)$.

Inclusion–exclusion:

$$\mathbb{P}\left(\bigcup_{i=1}^n E_i\right) = \sum_{r=1}^n (-1)^{r+1} \sum_{i_1 < \cdots < i_r} \mathbb{P}(E_{i_1} \cdots E_{i_r})$$

Adding singles over-counts overlaps; subtracting pairs over-removes triple overlaps; and so on, alternately.

Bounds (truncating alternately over/under-shoots):
 $\mathbb{P}(\bigcup E_i) \leq \sum \mathbb{P}(E_i)$ (Boole / union bound);
 $\mathbb{P}(\bigcup E_i) \geq \sum \mathbb{P}(E_i) - \sum_{i < j} \mathbb{P}(E_i E_j)$; next truncation is $\leq$ again.

Equally likely outcomes
 $|S| = N$ finite, all equally likely $\Rightarrow \mathbb{P}(E) = \frac{|E|}{|S|}$.

Matching (hat) problem
 N people pick hats at random, E_i = person i gets own hat.
 $\mathbb{P}(E_{i_1} \cdots E_{i_n}) = \frac{(N-n)!}{N!}$, $\binom{N}{n}$ such terms, so by incl.–excl.:

$$\mathbb{P}(\text{no match}) = \sum_{i=0}^N \frac{(-1)^i}{i!} \xrightarrow{N \rightarrow \infty} e^{-1} \approx 0.368$$

$\mathbb{P}(\text{exactly } k) = \frac{1}{k!} \sum_{i=0}^{N-k} \frac{(-1)^i}{i!} \approx \frac{e^{-1}}{k!}$, i.e. #matches $\approx$ Poisson(1).

3 Conditional Probability and Independence

Conditional probability
If $\mathbb{P}(F) > 0$: $\mathbb{P}(E \mid F) = \frac{\mathbb{P}(EF)}{\mathbb{P}(F)}$.

F becomes the new sample space: keep only outcomes in F and rescale so they sum to 1.

Multiplication rule:
$$\mathbb{P}(E_1 \cdots E_n) = \mathbb{P}(E_1) \mathbb{P}(E_2 | E_1) \mathbb{P}(E_3 | E_1 E_2) \cdots \mathbb{P}(E_n | E_1 \cdots E_{n-1})$$

- Urn 8R, 4W, 2 draws w/o repl., both red: $\frac{8}{12} \cdot \frac{7}{11} = \frac{14}{33}$.

- Deck into 4 piles of 13, each pile gets one ace: $\frac{39}{51} \cdot \frac{26}{50} \cdot \frac{13}{49} \approx 0.105$.
- Bridge: N+S hold 8 spades; E has exactly 3 of the other 5: $\frac{\binom{5}{3} \binom{21}{10}}{\binom{26}{13}} \approx 0.339$ (reduced sample space).

Law of total probability
 $\mathbb{P}(E) = \mathbb{P}(E|F)\mathbb{P}(F) + \mathbb{P}(E|F^c)(1 - \mathbb{P}(F))$.
General: $F_1, \dots, F_n$ a partition of S (disjoint, exhaustive):

$$\mathbb{P}(E) = \sum_{i=1}^n \mathbb{P}(E|F_i)\mathbb{P}(F_i)$$

$\mathbb{P}(E)$ is a weighted average of the conditional probabilities; condition on whatever makes the problem easy (often the first step).

Bayes’s formula
$$\mathbb{P}(F_j|E) = \frac{\mathbb{P}(E|F_j)\mathbb{P}(F_j)}{\sum_i \mathbb{P}(E|F_i)\mathbb{P}(F_i)}$$

Prior $\mathbb{P}(F_j) \rightarrow$ posterior $\mathbb{P}(F_j|E)$ after seeing evidence E .
Odds of A: $\frac{\mathbb{P}(A)}{\mathbb{P}(A^c)} = \frac{\mathbb{P}(A)}{1 - \mathbb{P}(A)}$ (“ α to 1”). Updating:

$$\frac{\mathbb{P}(H|E)}{\mathbb{P}(H^c|E)} = \frac{\mathbb{P}(H)}{\mathbb{P}(H^c)} \cdot \frac{\mathbb{P}(E|H)}{\mathbb{P}(E|H^c)}$$

new odds = old odds $\times$ likelihood ratio.
Diagnostic tests: sensitivity, specificity, prevalence
 D = has disease, $+$ / $-$ = test result.
Prevalence $\pi = \mathbb{P}(D)$ (prior, base rate)
Sensitivity $\mathbb{P}(+ \mid D)$ (true positive rate)
Specificity $\mathbb{P}(- \mid D^c)$ (true negative rate)
False pos. rate $\mathbb{P}(+ \mid D^c) = 1 - \text{spec}$
False neg. rate $\mathbb{P}(- \mid D) = 1 - \text{sens}$

Independence
 E, F independent $\iff \mathbb{P}(EF) = \mathbb{P}(E)\mathbb{P}(F)$ ($\iff \mathbb{P}(E|F) = \mathbb{P}(E)$ when $\mathbb{P}(F) > 0$).

- E, F indep. $\Rightarrow E, F^c$ indep. (and E^c, F^c).
- Three events** independent: all *four* conditions $\mathbb{P}(EFG) = \mathbb{P}(E)\mathbb{P}(F)\mathbb{P}(G)$ and each pair. Pairwise $\nRightarrow$ mutual (e.g. 2 dice: sum=7, die1=4, die2=3).
- If E, F, G indep., E is indep. of any event built from F, G (e.g. $F \cup G$).
- $E_1, \dots, E_n$ indep.: product rule for *every* subcollection. Infinite family: every finite subfamily.
- Disjoint $\neq$ independent! (Disjoint with positive probs $\Rightarrow$ dependent.)

Independent trials (success prob. p)

- $\mathbb{P}(\geq 1 \text{ success in } n) = 1 - (1 - p)^n$.
- $\mathbb{P}(\text{exactly } k \text{ in } n) = \binom{n}{k} p^k (1 - p)^{n-k}$.
- Parallel system works iff ≥ 1 works: $1 - \prod (1 - p_i)$.
- E before F** (E, F disjoint, repeated trials): $\frac{\mathbb{P}(E)}{\mathbb{P}(E) + \mathbb{P}(F)}$. Sum 5 before 7: $\frac{4}{4+6} = \frac{2}{5}$.
- Coupons (k collected, type i w.p. p_i): $\mathbb{P}(A_i) = 1 - (1 - p_i)^k$, $\mathbb{P}(A_i \cup A_j) = 1 - (1 - p_i - p_j)^k$.

Gambler’s ruin

A has i , B has $N - i$; A wins each round w.p. p , $q = 1 - p$. Condition on first toss: $P_i = pP_{i+1} + qP_{i-1}$, $P_0 = 0$, $P_N = 1$.

$$P_i = \begin{cases} \frac{1 - (q/p)^i}{1 - (q/p)^N}, & p \neq \frac{1}{2} \\ \frac{i}{N}, & p = \frac{1}{2} \end{cases}$$

$P_i + Q_i = 1$: game ends w.p. 1. ($i = 5, N = 15$: $p = .5 \rightarrow \frac{1}{3}$; $p = .6 \rightarrow \approx .87$.)

$\mathbb{P}(\cdot \mid F)$ is a probability

$Q(E) = \mathbb{P}(E|F)$ satisfies A1 to A3, so every result holds conditionally, e.g. $\mathbb{P}(E_1 \cup E_2|F) = \mathbb{P}(E_1|F) + \mathbb{P}(E_2|F) - \mathbb{P}(E_1E_2|F)$, and

$$\mathbb{P}(E|F) = \mathbb{P}(E|GF)\mathbb{P}(G|F) + \mathbb{P}(E|G^cF)\mathbb{P}(G^c|F)$$

Conditional independence given F : $\mathbb{P}(EG|F) = \mathbb{P}(E|F)\mathbb{P}(G|F) \iff \mathbb{P}(E|GF) = \mathbb{P}(E|F)$.

Cond. indep. $\nRightarrow$ indep. Two students, random easy/hard question: given the question they answer independently, but one getting it right suggests “easy”, raising the other’s chance. ($\mathbb{P}(E) = \mathbb{P}(G) = 0.7$, $\mathbb{P}(EG) = 0.53 \neq 0.49$.)

4 Random Variables

Basics

RV $X : S \rightarrow \mathbb{R}$. **pmf** $p(x) = \mathbb{P}\{X = x\}$, $p(x) \geq 0$, $\sum_x p(x) = 1$ (use this to find a constant c , e.g. $p(i) = c\lambda^i/i! \Rightarrow c = e^{-\lambda}$).
cdf $F(a) = \mathbb{P}\{X \leq a\} = \sum_{x \leq a} p(x)$: step function, jump $p(x_i)$ at x_i .

Expectation

$\mathbb{E}[X] = \sum_x x p(x)$ (centre of mass of the pmf).
Indicator I_A : $\mathbb{E}[I_A] = \mathbb{P}(A)$.
LOTUS: $\mathbb{E}[g(X)] = \sum_x g(x)p(x)$ (no need for the pmf of $g(X)$).
Linearity: $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$; $\mathbb{E}[\sum X_i] = \sum \mathbb{E}[X_i]$ (*always*, no independence needed).

*Counting trick: write $X = \sum I_i$ (indicators), then $\mathbb{E}[X] = \sum \mathbb{P}(A_i)$.
E.g. sum of n dice: $3.5n$; successes in trials with probs p_i : $\sum p_i$.*

Inspection paradox (buses of 36, 40, 44): random *bus* has mean 40; random *student’s* bus has mean $\frac{36^2+40^2+44^2}{120} \approx 40.27$ (big buses are sampled more).

Variance

$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$, $\text{SD} = \sqrt{\text{Var}}$.
 $\text{Var}(aX + b) = a^2\text{Var}(X)$ (shifting doesn’t change spread). Fair die: $\mathbb{E} = \frac{7}{2}$, $\text{Var} = \frac{35}{12}$.
Sum of indicators: $\mathbb{E}[X^2] = \sum_i \mathbb{P}(A_i) + \sum_{i \neq j} \mathbb{P}(A_i A_j)$, then subtract $(\mathbb{E}X)^2$.

Bernoulli and Binomial(n, p)

Bernoulli: $X \in \{0, 1\}$, $\mathbb{P}(X = 1) = p$; $\mathbb{E} = p$, $\text{Var} = p(1 - p)$.
Binomial = #successes in n indep. trials:

$$\mathbb{P}\{X = k\} = \binom{n}{k} p^k (1 - p)^{n-k}, \quad \mathbb{E} = np, \text{Var} = np(1 - p)$$

Ratio $\frac{\mathbb{P}\{X=k\}}{\mathbb{P}\{X=k-1\}} = \frac{(n-k+1)p}{k(1-p)}$: increases then decreases; **mode** at $\lfloor (n+1)p \rfloor$.

Poisson(λ)

$$\mathbb{P}\{X = i\} = e^{-\lambda} \frac{\lambda^i}{i!}, \quad i = 0, 1, \dots \quad \mathbb{E} = \text{Var} = \lambda$$

Approximates Bin(n, p) when n large, p small, $\lambda = np$ moderate (since $(1 - \lambda/n)^n \rightarrow e^{-\lambda}$).

Many independent rare chances $\Rightarrow$ Poisson: typos per page, 100-year-olds, wrong numbers dialled, α -particles, customers per day, hat matches ($\lambda = 1$).

Geometric(p)

X = #trials until the first success:

$$\mathbb{P}\{X = n\} = (1 - p)^{n-1} p, \quad n \geq 1; \quad \mathbb{P}\{X \geq k\} = (1 - p)^{k-1}$$

$$\mathbb{E}[X] = \frac{1}{p}, \quad \text{Var}(X) = \frac{1 - p}{p^2}. \quad (\text{Die until a 1: 6 rolls on average.})$$

Properties of the cdf

(1) F nondecreasing; (2) $\lim_{b \rightarrow \infty} F(b) = 1$; (3) $\lim_{b \rightarrow -\infty} F(b) = 0$; (4) right-continuous: $b_n \downarrow b \Rightarrow F(b_n) \rightarrow F(b)$.

$$\begin{aligned} \mathbb{P}\{a < X \leq b\} &= F(b) - F(a) \\ \mathbb{P}\{X < b\} &= F(b^-) = \lim_n F(b - \tfrac{1}{n}) \\ \mathbb{P}\{X = b\} &= F(b) - F(b^-) \text{ (jump size)} \\ \mathbb{P}\{X > b\} &= 1 - F(b), \quad \mathbb{P}\{X \geq b\} = 1 - F(b^-) \end{aligned}$$

Which model?

Fixed n trials, count successes Binomial(n, p)
Rare events, large n , count Poisson(np)
Wait for first success Geometric(p)
Draw w/o replacement counting / cond. prob.
“At least one” complement
Random walk to a boundary condition on step 1

Distribution summary

	pmf	$\mathbb{E}[X]$	$\text{Var}(X)$
Bern(p)	$p^x (1 - p)^{1-x}$	p	$p(1 - p)$
Bin(n, p)	$\binom{n}{k} p^k (1 - p)^{n-k}$	np	$np(1 - p)$
Poi(λ)	$e^{-\lambda} \lambda^k / k!$	λ	λ
Geom(p)	$(1 - p)^{k-1} p$	$1/p$	$(1 - p)/p^2$

Useful series

$$\begin{aligned} \sum_{k \geq 0} r^k &= \frac{1}{1-r}, \quad \sum_{k \geq 1} k r^{k-1} = \frac{1}{(1-r)^2} \quad (|r| < 1) \\ e^x &= \sum_{k \geq 0} \frac{x^k}{k!}, \quad (1 + \tfrac{x}{n})^n \rightarrow e^x \\ \sum_{k=1}^n k &= \frac{n(n+1)}{2}, \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6} \end{aligned}$$